Program: **Diploma in Civil and Environmental Engineering**	
Course Code: **6352B**	Course Title: **Green Buildings**
Semester: **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4(L:4 T:0 P:0)**	Periodspersemester:**60**

Course Objectives:

- To create awareness about the principles of green building technology
- To get a clear understanding of various renewable and non-renewable sources of energy

  along with their carbon foot prints
- To discuss about the energy efficient green building materials and to have understanding on the cost effective Building Technologies
- To get an idea about green building construction

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Explain the underlying principles and history of green building technology	15	Understanding
CO2	Compare various Renewable and Non-renewable sources of energy along with their carbon foot prints	14	Understanding
CO3	Illustrate the energy efficient green building materials	14	Understanding
CO4	Summarize the building codes and IGBC green building certification procedure.	15	Understanding
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		3
CO2	3				3		3
CO3	3				3		3
CO4	3				3		3

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the underlying principles and history of green building technology**		
M1.01	Define the term Green Building	3	Remembering
M1.02	Describe Green Building Rating Systems	4	Understanding
M1.03	Explain Environmental design (ED) strategies for building construction	4	Understanding
M1.04	Summarize green building materials and products	4	Understanding

Contents:

Introduction to Green Buildings. Definition of Green Buildings-Typical features of green buildings, Necessity, Initiatives, Green buildings in India.

Green Building Rating Systems (BREEAM,USGBC,LEED,IGBC,TERI-GRIHA, GREEN STAR),Criteria for rating, Energy efficient criteria ,environmental benefits economic benefits, health and social benefits , Major energy efficiency areas for building, contribution of buildings towards Global Warming.

Environmental design (ED) strategies for building construction - Process: Improvement in environmental quality in civil structure.

Green building materials and products-Bamboo, Rice husk ash concrete, plastic bricks, Bagasse particle board, Insulated concrete forms. reuse of waste material-Plastic, rubber, Newspaper, wood, Nontoxic paint- Green roofing.

<table>
<tr><td>CO2</td><td colspan="3">Compare various Renewable and Non-renewable sources of energy along with their carbon foot prints</td></tr>
<tr><td>M2.01</td><td>List Renewable and Non-renewable sources of energy</td><td>3</td><td>Remembering</td></tr>
<tr><td>M2.02</td><td>Explain global warming and global efforts to reduce carbon emissions</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Define Energy efficiency in Green buildings</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Describe Energy auditing.</td><td>4</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test I</td><td>1</td><td></td></tr>
</table>

Contents:

Sources of Energy: Renewable and Non-renewable sources of energy; Coal, Petroleum, Nuclear, Wind, Solar, Hydro, Geothermal sources; potential of these sources, hazards, pollution.

Global Warming – Definition - Causes and Effects - Contribution of Buildings towards global Warming - Carbon Footprint – Global Efforts to reduce carbon Emissions.

Major Energy efficient areas for buildings – Embodied Energy in Materials-Green Materials - Comparison of Initial cost of Green V/s Conventional Building - Life cycle cost of Buildings.

Energy Auditing: Low energy design, hybrid systems, modeling and simulation of energy systems, integration of PV and wind systems in the building, wind solar and other non-conventional energy systems, solar thermal applications for heating and cooling, electricity generation in buildings.

<table>
<tr><td>CO3</td><td colspan="3">Illustrate the energy efficient green building materials</td></tr>
<tr><td>M3.01</td><td>Define Green composites for buildings.</td><td>4</td><td>Remembering</td></tr>
<tr><td>M3.02</td><td>Listing of Renewable and Indigenous Building Materials.</td><td>4</td><td>Remembering</td></tr>
<tr><td>M3.03</td><td>Describe the Green Building Planning and Specifications.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Explain Waste & Water management and Recycling in Sustainable Facilities.</td><td>3</td><td>Understanding</td></tr>
</table>

Contents:

Green Composites for Buildings-Concepts of Green Composites. Urban Environment and Green Buildings. Green Cover and Built Environment.

Renewable and recyclable resources; energy efficient materials; Green cement, Biodegradable materials, Smart materials, Manufactured Materials, Volatile Organic Compounds (VOC’s), Natural Non-Petroleum Based Materials, Recycled materials, Renewable and Indigenous Building Materials- Engineering evaluation of these materials. Green Building Planning and Specifications: Environment friendly and cost-effective Building Technologies, Integrated Life cycle design of Materials and Structures, Green Strategies for Building Systems, Alternative Construction Methods, Energy Conservation Measures in Buildings.

Waste & Water management and Recycling in Sustainable Facilities, Heating, Ventilation and Air Conditioning, Passive Solar & Daylight, Plumbing and its Effect on Energy Consumption

CO4	**Describe the construction of green building**		
M4.01	Identify the components of Green building	3	Understanding
M4.02	Summarize Energy efficient construction	4	Understanding
M4.03	Describe the materials and construction techniques	4	Understanding
M4.04	Explain Cost/benefit analysis of green buildings	4	Understanding
	Series Test II	1	

Contents:

Design Features of Green Building. Components/features of Green Building, Site selection, Energy Efficiency, Water efficiency, Material Efficiency, Indoor Air Quality - Site selection strategies, Landscaping, building form, orientation, building envelope and fenestration - material and construction techniques, roofs, walls, fenestration and shaded finishes, advanced passive heating and cooling techniques, waste reduction during construction. Cost/benefit analysis of green buildings, Life-cycle analysis of green buildings, Case studies of rated buildings (new and existing)

Text/Reference:

T/R	Book Title/Author
T1	Jagadish. K.S. Alternative Building Materials and Technologies, New age International Pvt.Ltd Publishers, 2008
T2	Ross SpiegelG, Green Building Materials A Guide to Product Selection and Specification,3rd Edition by, John Wiley &Sons, 2010
T3	Traci Rose Rider, Stacy Glass, Jessica McNaughton, Understanding Green BuildingMaterials, W.W.Norton andCompany, 2011
T4	Automation Systems in Smart and Green Buildings (Modern Building Technology), Er. VK Jain, Khanna Publishers
T5	Integrated Life Cycle Design of Structures – By AskoSarja – SPON Press
T6	Non-conventional Energy Resources – By D S Chauhan and S K Sreevasthava – New Age International Publishers
T7	Emerald Architecture: case studies in green buildings, The Magazine of Sustainable Design
R1	Sustainable Building - Design Manual Pt 1 & 2, The Energy and Resources Institute,TERI, 2004
R2	BIM Handbook: A Guide to Building Information Modeling for Owners, Managers,Designers, Engineers and Contractors- Chuck Eastman, et al.
R3	BIS, National Building Code 2005, New Delhi, 2005
R4	Energy Conservation Building Code of India, User manual, 2007
R5	Sustainable Building Technical Manual - Green Building Practices for Design, Construction and Operations, US Green Building Council, 2011.
R6	P.K. Singh, Rainwater Harvesting: Low cost indigenous and innovative technologies,Macmillan Publishers India, 2008

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	https://www.coursera.org/
3	https://swayam.gov.in/